THOMSON COURSE TECHNOLOGY

Guide to Network Defense and Countermeasures Third Edition

Chapter 3

Network Traffic Signatures

Objectives

- Explain the Common Vulnerabilities and Exposures (CVE) standard
- Describe the concepts of signature analysis
- Detect normal and suspicious traffic signatures
- Identify suspicious events

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Examining the Common Vulnerabilities and Exposures Standard

- To prevent attacks, make sure your security devices share information and coordinate with one another
  - Each device uses its own “language”
  - The way they interpret signatures might differ
- Common Vulnerabilities and Exposures (CVE) standard
  - Enables devices to share information using the same standard

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How the CVE Works, Part 1

- CVE enables hardware and security devices to draw from the same database of vulnerabilities
- Benefits
  - Stronger security
  - Better performance
- When purchasing an intrusion detection and prevention system (IDPS)
  - Make sure they support CVE

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How the CVE Works, Part 2

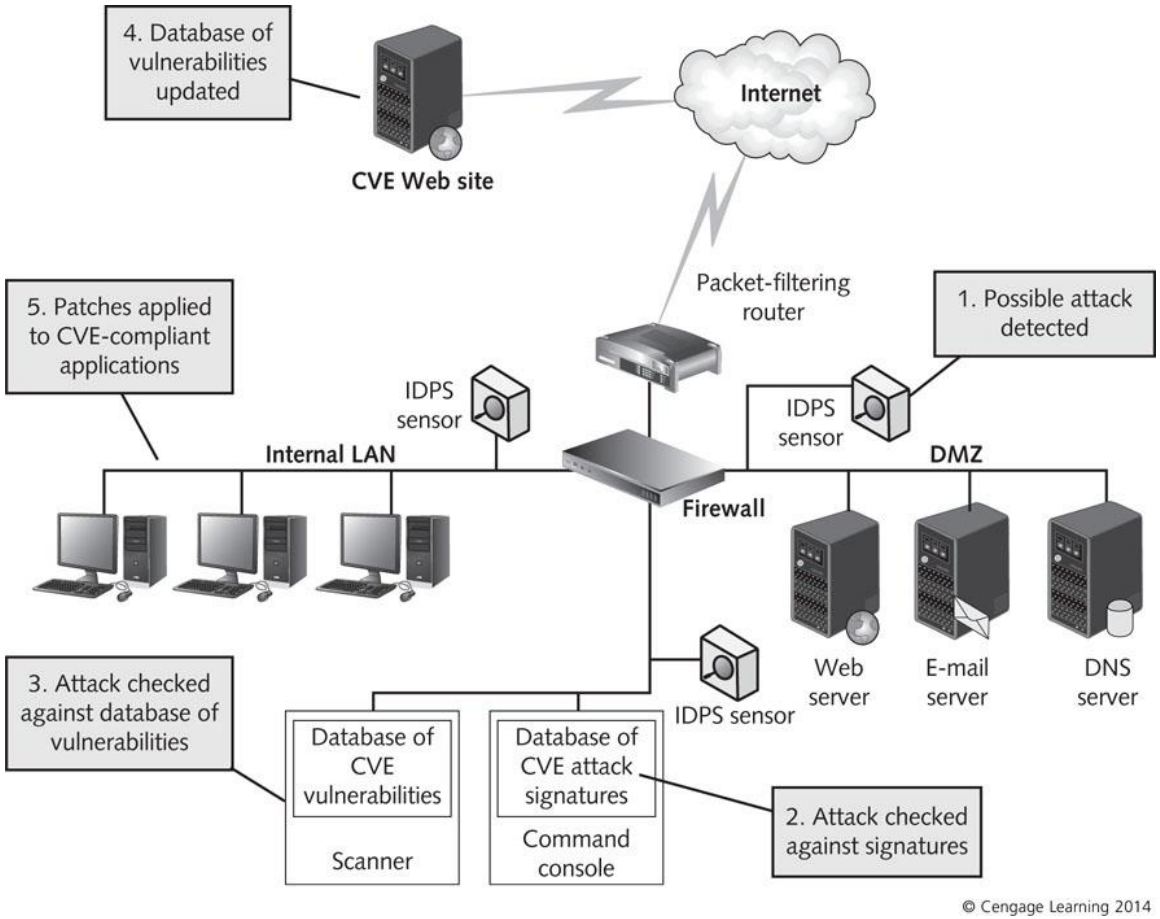

Figure 3-1 CVE enables multiple devices to work together to detect possible attacks

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Scanning CVE Vulnerabilities Descriptions

- View current CVE vulnerabilities online
  - List can be downloaded
- The CVE list is not a vulnerability database that can be used to repair attacks on an IDPS
- Information in a CVE reference
  - Name of the vulnerability
  - Short description
  - References to the event in other databases
    - Such as BUGTRAQ

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CVE Candidate Listing

CVE-ID	
CVE-2012-0390 (under review)	Learn more at National Vulnerability Database (NVD) • Severity Rating • Fix Information • Vulnerable Software Versions • SCAP Mappings
Description	
The DTLS implementation in GnuTLS 3.0.10 and earlier executes certain error-handling code only if there is a specific relationship between a padding length and the ciphertext size, which makes it easier for remote attackers to recover partial plaintext via a timing side-channel attack, a related issue to CVE-2011-4108.	
References	
Note: References are provided for the convenience of the reader to help distinguish between vulnerabilities. The list is not intended to be complete.	
• MISC:http://www.isg.rhul.ac.uk/~kp/dtls.pdf	
Status	
Candidate	This CVE Identifier has "Candidate" status and must be reviewed and accepted by the CVE Editorial Board before it can be updated to official "Entry" status on the CVE List. It may be modified or even rejected in the future.
Phase	
Assigned (20120105)	
Votes	
Comments	

Source: www.cve.mitre.org

Figure 3-2 CVE candidate listing CVE-2012-0390

Understanding Signature Analysis, Part 1

- Signature – set of characteristics used to define a type of network activity
  - IP numbers and options, TCP flags, and port numbers are examples
- Some intrusion-detection devices assemble databases of “normal” traffic signatures
  - Deviations from normal signatures trigger an alarm
- Other devices refer to a database of well-known attack signatures
  - Traffic that matches stored signatures triggers an alarm

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Understanding Signature Analysis, Part 2

- Signature analysis:
  - Practice of analyzing and understanding TCP/IP communications to determine whether they are legitimate or suspicious
- Bad header information
  - Packets are often altered through header information
  - Suspicious signatures can include malformed
    - Source and destination IP address
    - Source and destination port number
    - IP options, protocol and checksums
    - IP fragmentation flags, offset, or identification

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Understanding Signature Analysis, Part 3

- Bad header information
  - Checksum
    - Simple error-checking procedure
    - Determines whether a message has been damaged or tampered with while in transit
    - Uses a mathematical formula
- Suspicious data payload
  - Payload
    - Actual data sent from an application on one computer to an application on another
  - Some IDPSs check for specific strings in the payload

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Understanding Signature Analysis, Part 4

- Suspicious data payload (cont'd)
  - Remote-access Trojans (RATs):open back doors that give the remote attacker administrative rights
  - Unix Sendmail program is exploited by adding codes to packet contents
- Single-Packet Attacks
  - Also called "atomic attacks"
  - Completed by sending a single network packet from client to host
  - Does not need a connection to be established
  - Changes to IP option settings can cause a server to freeze up

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IP Options Settings

Option Number	Name of option
0	End of Options
1	No operation
2	Security
3	Loose Source and Record Routing
4	Internet Timestamp
7	Record Return Route
8	Option has been deprecated
9	Strict Source and Record Routing

Table 3-1 IP options settings

Understanding Signature Analysis, Part 5

- Multiple-Packet Attacks
  - Also called “composite attacks”
  - Require a series of packets to be received and executed for the attack to be completed
  - Especially difficult to detect
  - Denial-of-service (DoS) attacks are obvious examples
    - ICMP flood: a type of DoS attack that occurs when multiple ICMP packets are sent to a single host on a network
      - Server becomes so busy responding to ICMP requests that it cannot process other traffic

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Analyzing Packets

- Packet sniffer
  - Captures information about each TCP/IP packet it detects
  - Capturing packets and studying them can help you better understand what makes up a signature
  - Example:
    - Wireshark
  - Be familiar with elements of TCP/IP packets discussed on pages 86-88 of textbook

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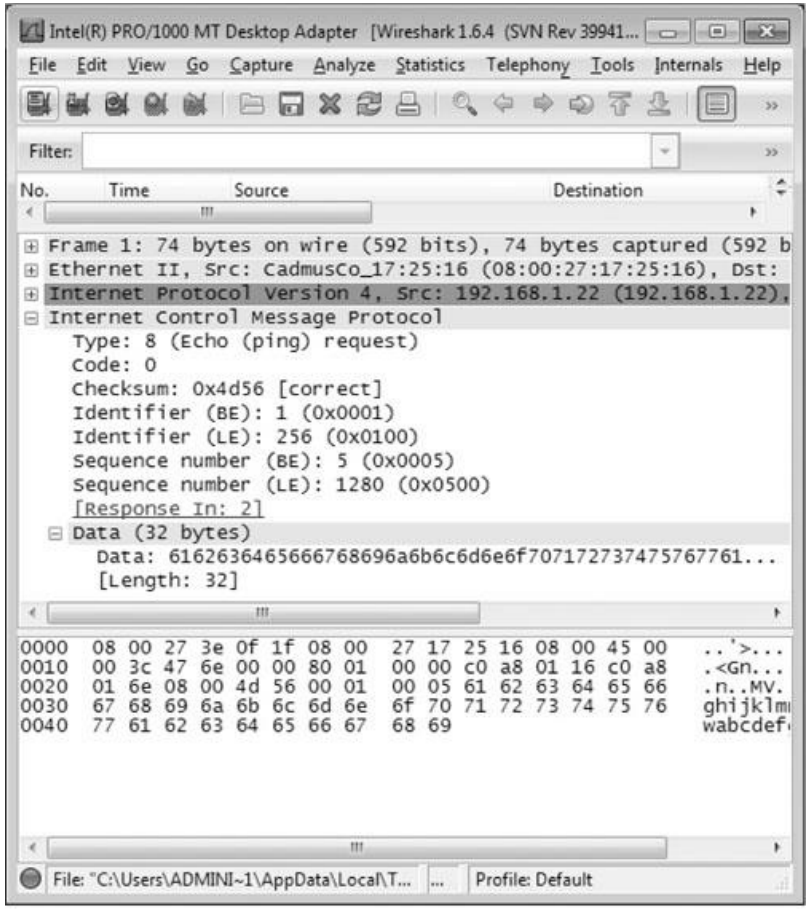
An ICMP Echo Request Packet Capture



Source: Wireshark

Figure 3-3 An ICMP echo request packet capture

Analyzing Traffic Signatures

- Need to detect whether traffic is normal or suspicious
- Network baselining
  - Process of determining what is normal for your network before you can identify anomalies

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Examining Normal Network Traffic Signatures, Part 1

- Important TCP flags
  - SYN (0x2) – *synchronize flag* is sent when a connection is initiated
  - ACK (0x10) – *acknowledgement flag* is set to signal that the previous packet was received
  - PSH (0x8) – *push flag* indicates that immediate delivery is required
  - URG (0x20) – *urgent flag* is used when urgent data is being sent
  - RST (0x4) – *reset flag* is sent when one computer wants to stop and restart the connection in response to a problem

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Examining Normal Network Traffic Signatures, Part 2

- Important TCP flags (continued)
  - FIN (0x1) – *finished flag* lets one computer know that the other is finished sending data
- Placement and use of these flags are definite
  - Deviations from normal use mean that the communication is suspicious

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TShark Capture of a TCP Stream

	Seconds from start of capture	Source IP address		Destination IP address	Protocol	Size	Source Port		Destination Port	TCP flags	Sequence/Acknowledgement numbers, etc.
1	18.743157	192.168.1.132	->	192.168.1.1	TCP	66	49263	>	1780	[SYN]	Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
2	18.748737	192.168.1.1	->	192.168.1.132	TCP	66	1780	>	49263	[SYN, ACK]	Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM=1 WS=2
3	18.748920	192.168.1.132	->	192.168.1.1	TCP	54	49263	>	1780	[ACK]	Seq=1 Ack=1 Win=65700 Len=0
4	18.749473	192.168.1.132	->	192.168.1.1	TCP	243	49263	>	1780	[PSH, ACK]	Seq=1 Ack=1 Win=65700 Len=189
5	18.758879	192.168.1.1	->	192.168.1.132	TCP	60	1780	>	49263	[ACK]	Seq=1 Ack=190 Win=5840 Len=0
6	18.759032	192.168.1.1	->	192.168.1.132	TCP	156	1780	>	49263	[PSH, ACK]	Seq=1 Ack=190 Win=5840 Len=102
7	18.760956	192.168.1.1	->	192.168.1.132	TCP	1514	1780	>	49263	[ACK]	Seq=103 Ack=190 Win=5840 Len=1460
8	18.761105	192.168.1.132	->	192.168.1.1	TCP	54	49263	>	1780	[ACK]	Seq=190 Ack=1563 Win=65700 Len=0
9	18.775169	192.168.1.1	->	192.168.1.132	TCP	1514	1780	>	49263	[ACK]	Seq=1563 Ack=190 Win=5840 Len=1460
10	18.777046	192.168.1.1	->	192.168.1.132	TCP	1514	1780	>	49263	[ACK]	Seq=3023 Ack=190 Win=5840 Len=1460
11	18.777213	192.168.1.132	->	192.168.1.1	TCP	54	49263	>	1780	[ACK]	Seq=190 Ack=4483 Win=65700 Len=0
12	18.777292	192.168.1.1	->	192.168.1.132	TCP	122	1780	>	49263	[FIN, PSH, ACK]	Seq=4483 Ack=190 Win=5840 Len=68
13	18.777446	192.168.1.132	->	192.168.1.1	TCP	54	49263	>	1780	[ACK]	Seq=190 Ack=4552 Win=65632 Len=0
14	18.778399	192.168.1.132	->	192.168.1.1	TCP	54	49263	>	1780	[FIN, ACK]	Seq=190 Ack=4552 Win=65632 Len=0
15	18.783569	192.168.1.1	->	192.168.1.132	TCP	60	1780	>	49263	[ACK]	Seq=4552 Ack=191 Win=5840 Len=0

Source: Wireshark

Figure 3-6 TShark capture of a TCP stream

Examining Normal Network Traffic Signatures, Part 3

- FTP Signatures
  - Organizations that operate a public FTP server should regularly review the signatures of packets that attempt to access that server
  - Normal connection signature includes a three-way handshake
  - The sequence of packets is shown in the next slides

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The Beginning of an FTP Session

```
| 0.000000 192.168.1.132 -> 192.168.1.110 TCP 66 50580 > 21 [SYN] Seq=0 Win=8192 Len=0 MSS=1460

0000  08 00 27 3e 0f 1f 08 00 27 17 25 16 08 00 45 00   ..'>....'.%...E.
0010  00 34 19 de 40 00 80 06 00 00 c0 a8 01 84 c0 a8   .4..@...........
0020  01 6e c5 94 00 15 54 9d 21 a7 00 00 00 00 80 02   .n....T.!.......
0030  20 00 84 69 00 00 02 04 05 b4 01 03 03 00 01 01    ..i............
0040  04 02                                             ..

  0.000533 192.168.1.110 -> 192.168.1.132 TCP 66 21 > 50580 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len

0000  08 00 27 17 25 16 08 00 27 3e 0f 1f 08 00 45 00   ..'.%...'>....E.
0010  00 34 24 af 40 00 80 06 51 d2 c0 a8 01 6e c0 a8   .4$.@...Q....n..
0020  01 84 00 15 c5 94 44 6e f5 cb 54 9d 21 a8 80 12   ......Dn..T.!...
0030  20 00 54 94 00 00 02 04 05 b4 01 03 03 08 01 01    .T.............
0040  04 02                                             ..

  0.000648 192.168.1.132 -> 192.168.1.110 TCP 54 50580 > 21 [ACK] Seq=1 Ack=1 Win=8192 Len=0

0000  08 00 27 3e 0f 1f 08 00 27 17 25 16 08 00 45 00   ..'>....'.%...E.
0010  00 28 19 df 40 00 80 06 00 00 c0 a8 01 84 c0 a8   .(..@...........
0020  01 6e c5 94 00 15 54 9d 21 a8 44 6e f5 cc 50 10   .n....T.!.Dn..P.
0030  20 00 84 5d 00 00                                  ..]..

  0.001455 192.168.1.110 -> 192.168.1.132 FTP 81 Response: 220 Microsoft FTP Service

0000  08 00 27 17 25 16 08 00 27 3e 0f 1f 08 00 45 00   ..'.%...'>....E.
0010  00 43 24 b0 40 00 80 06 51 c2 c0 a8 01 6e c0 a8   .C$.@...Q....n..
0020  01 84 00 15 c5 94 44 6e f5 cc 54 9d 21 a8 50 18   ......Dn..T.!.P.
0030  01 00 19 5b 00 00 32 32 30 20 4d 69 63 72 6f 73   ...[..220 Micros
0040  6f 66 74 20 46 54 50 20 53 65 72 76 69 63 65 0d   oft FTP Service.
0050  0a                                                .

  0.208118 192.168.1.132 -> 192.168.1.110 TCP 54 50580 > 21 [ACK] Seq=1 Ack=28 Win=8165 Len=0

0000  08 00 27 3e 0f 1f 08 00 27 17 25 16 08 00 45 00   ..'>....'.%...E.
0010  00 28 19 e0 40 00 80 06 00 00 c0 a8 01 84 c0 a8   .(..@...........
0020  01 6e c5 94 00 15 54 9d 21 a8 44 6e f5 e7 50 10   .n....T.!.Dn..P.
0030  1f e5 84 5d 00 00                                  ...]..
```

Source: Wireshark

Figure 3-7 The beginning of an FTP session

Continuation of an FTP Session

```
  2.123399 192.168.1.132 -> 192.168.1.110 FTP 65 Request: USER dean

0000  08 00 27 3e 0f 1f 08 00 27 17 25 16 08 00 45 00   ..'>....'.%...E.
0010  00 33 19 e1 40 00 80 06 00 00 c0 a8 01 84 c0 a8   .3..@...........
0020  01 6e c5 94 00 15 54 9d 21 a8 44 6e f5 e7 50 18   .n....T.!.Dn..P.
0030  1f e5 84 68 00 00 55 53 45 52 20 64 65 61 6e 0d   ...h..USER dean.
0040  0a                                                .

  2.124041 192.168.1.110 -> 192.168.1.132 FTP 87 Response: 331 Password required for dean.

0000  08 00 27 17 25 16 08 00 27 3e 0f 1f 08 00 45 00   ..'.%...'>....E.
0010  00 49 24 b1 40 00 80 06 51 bb c0 a8 01 6e c0 a8   .I$.@...Q....n..
0020  01 84 00 15 c5 94 44 6e f5 e7 54 9d 21 b3 50 18   ......Dn..T.!.P.
0030  01 00 f9 d8 00 00 33 33 31 20 50 61 73 73 77 6f   ......331 Passwo
0040  72 64 20 72 65 71 75 69 72 65 64 20 66 6f 72 20   rd required for
0050  64 65 61 6e 2e 0d 0a                              dean...

  2.317076 192.168.1.132 -> 192.168.1.110 TCP 54 50580 > 21 [ACK] Seq=12 Ack=61 Win=8132 Le

0000  08 00 27 3e 0f 1f 08 00 27 17 25 16 08 00 45 00   ..'>....'.%...E.
0010  00 28 19 e2 40 00 80 06 00 00 c0 a8 01 84 c0 a8   .(..@...........
0020  01 6e c5 94 00 15 54 9d 21 b3 44 6e f6 08 50 10   .n....T.!.Dn..P.
0030  1f c4 84 5d 00 00                                 ...]..

  6.426319 192.168.1.132 -> 192.168.1.110 FTP 69 Request: PASS Pa$$word

0000  08 00 27 3e 0f 1f 08 00 27 17 25 16 08 00 45 00   ..'>....'.%...E.
0010  00 37 19 e3 40 00 80 06 00 00 c0 a8 01 84 c0 a8   .7..@...........
0020  01 6e c5 94 00 15 54 9d 21 b3 44 6e f6 08 50 18   .n....T.!.Dn..P.
0030  1f c4 84 6c 00 00 50 41 53 53 20 50 61 24 24 77   ...l..PASS Pa$$w
0040  6f 72 64 0d 0a                                    ord..

  6.427227 192.168.1.110 -> 192.168.1.132 FTP 75 Response: 230 User logged in.

0000  08 00 27 17 25 16 08 00 27 3e 0f 1f 08 00 45 00   ..'.%...'>....E.
0010  00 3d 24 b2 40 00 80 06 51 c6 c0 a8 01 6e c0 a8   .=$.@...Q....n..
0020  01 84 00 15 c5 94 44 6e f6 08 54 9d 21 c2 50 18   ......Dn..T.!.P.
0030  01 00 99 e6 00 00 32 33 30 20 55 73 65 72 20 6c   ......230 User l
0040  6f 67 67 65 64 20 69 6e 2e 0d 0a                  ogged in...
```

Source: Wireshark

Figure 3-8 Continuation of an FTP session

The Teardown of an FTP Data Connection

```
15.632892 192.168.1.110 -> 192.168.1.132 TCP 60 21 > 50580 [FIN, ACK] Seq=312 Ack=113

0000  08 00 27 17 25 16 08 00 27 3e 0f 1f 08 00 45 00   ..'.%...'>....E.
0010  00 28 24 c4 40 00 80 06 51 c9 c0 a8 01 6e c0 a8   .($.@...Q....n..
0020  01 84 00 15 c5 94 44 6e f7 03 54 9d 22 18 50 11   ......Dn..T.".P.
0030  01 00 b2 bf 00 00 00 00 00 00 00 00               ............

15.632960 192.168.1.132 -> 192.168.1.110 TCP 54 50580 > 21 [ACK] Seq=113 Ack=313 Win=7

0000  08 00 27 3e 0f 1f 08 00 27 17 25 16 08 00 45 00   ..'>....'.%...E.
0010  00 28 19 f4 40 00 80 06 00 00 c0 a8 01 84 c0 a8   .(..@...........
0020  01 6e c5 94 00 15 54 9d 22 18 44 6e f7 04 50 10   .n....T.".Dn..P.
0030  1e c9 84 5d 00 00                                 ...]..

15.635862 192.168.1.132 -> 192.168.1.110 TCP 54 50580 > 21 [FIN, ACK] Seq=113 Ack=313

0000  08 00 27 3e 0f 1f 08 00 27 17 25 16 08 00 45 00   ..'>....'.%...E.
0010  00 28 19 f5 40 00 80 06 00 00 c0 a8 01 84 c0 a8   .(..@...........
0020  01 6e c5 94 00 15 54 9d 22 18 44 6e f7 04 50 11   .n....T.".Dn..P.
0030  1e c9 84 5d 00 00                                 ...]..

15.636335 192.168.1.110 -> 192.168.1.132 TCP 60 21 > 50580 [ACK] Seq=313 Ack=114 Win=6

0000  08 00 27 17 25 16 08 00 27 3e 0f 1f 08 00 45 00   ..'.%...'>....E.
0010  00 28 24 c5 40 00 80 06 51 c8 c0 a8 01 6e c0 a8   .($.@...Q....n..
0020  01 84 00 15 c5 94 44 6e f7 04 54 9d 22 19 50 10   ......Dn..T.".P.
0030  01 00 b2 be 00 00 00 00 00 00 00 00               ............
```

Source: Wireshark

Figure 3-9 The teardown of an FTP data connection

Examining Normal Network Traffic Signatures, Part 4

- Web Signatures
  - Most of the signatures in log files are Web related
  - When a signature is Web-related:
    - It consists of packets sent back and forth from a Web browser to a Web server as a connection is made
  - Normal communication consists of a sequence of packets distinguished by their TCP flags

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A Normal Exchange of Packets between a Web Browser and Server

```
  1.825108 192.168.1.132 -> 65.54.87.10  TCP 66 50969 > 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 :

0000  58 6d 8f 04 8a d3 08 00 27 17 25 16 08 00 45 00   Xm......'.%...E.
0010  00 34 25 71 40 00 80 06 00 00 c0 a8 01 84 41 36   .4%q@.........A6
0020  57 0a c7 19 00 50 1a f0 b1 44 00 00 00 00 80 02   W....P...D......
0030  20 00 5a 93 00 00 02 04 05 b4 01 03 03 02 01 01    .Z.............
0040  04 02                                             ..

  1.849810 58:6d:8f:04:8a:d3 -> ff:ff:ff:ff:ff:ff ARP 60 who has 192.168.1.132?  Tell 192.168.1.1

0000  ff ff ff ff ff ff 58 6d 8f 04 8a d3 08 06 00 01   ......Xm........
0010  08 00 06 04 00 01 58 6d 8f 04 8a d3 c0 a8 01 01   ......Xm........
0020  00 00 00 00 00 00 c0 a8 01 84 00 00 00 00 00 00   ................
0030  00 00 00 00 00 00 00 00 10 41 1e ff               .........A..

  1.849838 08:00:27:17:25:16 -> 58:6d:8f:04:8a:d3 ARP 42 192.168.1.132 is at 08:00:27:17:25:16

0000  58 6d 8f 04 8a d3 08 00 27 17 25 16 08 06 00 01   Xm......'.%.....
0010  08 00 06 04 00 02 08 00 27 17 25 16 c0 a8 01 84   ........'.%.....
0020  58 6d 8f 04 8a d3 c0 a8 01 01                     Xm........

  1.852508  65.54.87.10 -> 192.168.1.132 TCP 66 80 > 50968 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MS:

0000  08 00 27 17 25 16 58 6d 8f 04 8a d3 08 00 45 20   ..'.%.Xm......E
0010  00 34 5d 66 40 00 36 06 8c d1 41 36 57 0a c0 a8   .4]f@.6...A6W...
0020  01 84 00 50 c7 18 9c 36 4c e6 2b 07 39 c7 80 12   ...P...6L.+.9...
0030  ff ff 00 45 00 00 02 04 05 b4 01 03 03 04 04 02   ...E............
0040  00 00                                             ..

  1.852745 192.168.1.132 -> 65.54.87.10  TCP 54 50968 > 80 [ACK] Seq=1 Ack=1 Win=65700 Len=0

0000  58 6d 8f 04 8a d3 08 00 27 17 25 16 08 00 45 00   Xm......'.%...E.
0010  00 28 25 72 40 00 80 06 00 00 c0 a8 01 84 41 36   .(%r@.........A6
0020  57 0a c7 18 00 50 2b 07 39 c7 9c 36 4c e7 50 10   W....P+.9..6L.P.
0030  40 29 5a 87 00 00                                 @)Z...
```

Source: Wireshark

Figure 3-10 A normal exchange of packets between a Web browser and a Web server

Examining Normal Network Traffic Signatures, Part 5

- Web Signatures (cont'd)
  - Once the handshake is complete:
    - Web browser sends a request to the Web server for Web page data (called an HTTP GET packet)

```
1.855278 192.168.1.132 -> 65.54.87.10  HTTP 399 GET /i/52/79D26BA7D3D2DF25DBDD991B35E16.jpg HTTP/1.1

0000  58 6d 8f 04 8a d3 08 00 27 17 25 16 08 00 45 00   Xm......'.%...E.
0010  01 81 25 75 40 00 80 06 00 00 c0 a8 01 84 41 36   ..%u@.........A6
0020  57 0a c7 17 00 50 a7 1c 46 4f d3 06 5e 67 50 18   W....P..FO..^gP.
0030  40 29 5b e0 00 00 47 45 54 20 2f 69 2f 35 32 2f   @)[...GET /i/52/
0040  37 39 44 32 36 42 41 37 44 33 44 32 44 46 32 35   79D26BA7D3D2DF25
0050  44 42 44 44 39 39 31 42 33 35 45 31 36 2e 6a 70   DBDD991B35E16.jp
0060  67 20 48 54 54 50 2f 31 2e 31 0d 0a 41 63 63 65   g HTTP/1.1..Acce
0070  70 74 3a 20 69 6d 61 67 65 2f 70 6e 67 2c 20 69   pt: image/png, i
0080  6d 61 67 65 2f 73 76 67 2b 78 6d 6c 2c 20 69 6d   mage/svg+xml, im
0090  61 67 65 2f 2a 3b 71 3d 30 2e 38 2c 20 2a 2f 2a   age/*;q=0.8, */*
00a0  3b 71 3d 30 2e 35 0d 0a 52 65 66 65 72 65 72 3a   ;q=0.5..Referer:
00b0  20 68 74 74 70 3a 2f 2f 77 77 77 2e 6d 73 6e 2e    http://www.msn.
00c0  63 6f 6d 2f 3f 6f 63 69 64 3d 69 65 68 70 0d 0a   com/?ocid=iehp..
00d0  41 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 65 3a   Accept-Language:
00e0  20 65 6e 2d 55 53 0d 0a 55 73 65 72 2d 41 67 65    en-US..User-Age
00f0  6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20   nt: Mozilla/5.0 
0100  28 63 6f 6d 70 61 74 69 62 6c 65 3b 20 4d 53 49   (compatible; MSI
0110  45 20 39 2e 30 3b 20 57 69 6e 64 6f 77 73 20 4e   E 9.0; Windows N
0120  54 20 36 2e 31 3b 20 57 4f 57 36 34 3b 20 54 72   T 6.1; WOW64; Tr
0130  69 64 65 6e 74 2f 35 2e 30 29 0d 0a 41 63 63 65   ident/5.0)..Acce
0140  70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 7a 69   pt-Encoding: gzi
0150  70 2c 20 64 65 66 6c 61 74 65 0d 0a 48 6f 73 74   p, deflate..Host
0160  3a 20 63 6f 6c 2e 73 74 62 2e 73 2d 6d 73 6e 2e   : col.stb.s-msn.
0170  63 6f 6d 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a   com..Connection:
0180  20 4b 65 65 70 2d 41 6c 69 76 65 0d 0a 0d 0a       Keep-Alive....
```

Source: Wireshark

Figure 3-11 An HTTP GET packet

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Examining Abnormal Network Traffic Signatures

- Categories
  - Informational
    - Traffic might not be malicious but could be used to verify whether an attack has been successful
  - Reconnaissance
    - Attacker's attempt to gain information
  - Unauthorized access
    - Traffic caused by someone who has gained unauthorized access
  - Denial of service
    - Traffic might be part of an attempt to slow or halt all connections on a network device

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Examining Abnormal Network Traffic Signatures, Part 2

- Ping Sweeps
  - Also called an ICMP sweep
  - Used by attackers to determine the location of a host
  - Attacker sends a series of ICMP echo request packets in a range of IP addresses
  - Ping sweep alone does not cause harm
    - IP address used in the ping sweep should be noted in order to track further activity
    - AN IDPS could be configured to transmit an alarm and block transmissions if this IP address attempts to connect to a specific host on a network

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An Automated Ping Sweep

```
2.717757 192.168.1.110 -> 67.9.155.1  ICMP 42 Echo (ping) request  id=0x9188, seq=0/0, ttl=39
2.717914 192.168.1.110 -> 67.9.155.2  ICMP 42 Echo (ping) request  id=0xab94, seq=0/0, ttl=42
2.718227 192.168.1.110 -> 67.9.155.3  ICMP 42 Echo (ping) request  id=0x0ef3, seq=0/0, ttl=45
2.718309 192.168.1.110 -> 67.9.155.4  ICMP 42 Echo (ping) request  id=0x08e6, seq=0/0, ttl=50
2.718402 192.168.1.110 -> 67.9.155.5  ICMP 42 Echo (ping) request  id=0x50c8, seq=0/0, ttl=53
2.718487 192.168.1.110 -> 67.9.155.6  ICMP 42 Echo (ping) request  id=0xddc7, seq=0/0, ttl=48
2.718580 192.168.1.110 -> 67.9.155.7  ICMP 42 Echo (ping) request  id=0xb78a, seq=0/0, ttl=38
2.718654 192.168.1.110 -> 67.9.155.8  ICMP 42 Echo (ping) request  id=0xbcd2, seq=0/0, ttl=50
2.718723 192.168.1.110 -> 67.9.155.9  ICMP 42 Echo (ping) request  id=0xb453, seq=0/0, ttl=52
2.718792 192.168.1.110 -> 67.9.155.10 ICMP 42 Echo (ping) request  id=0xd793, seq=0/0, ttl=52
2.719075 192.168.1.110 -> 67.9.155.11 ICMP 42 Echo (ping) request  id=0xdc11, seq=0/0, ttl=39
2.719166 192.168.1.110 -> 67.9.155.12 ICMP 42 Echo (ping) request  id=0xf91c, seq=0/0, ttl=45
3.730437 192.168.1.110 -> 67.9.155.13 ICMP 42 Echo (ping) request  id=0xb7b9, seq=0/0, ttl=52
3.730716 192.168.1.110 -> 67.9.155.14 ICMP 42 Echo (ping) request  id=0x9f11, seq=0/0, ttl=53
3.730919 192.168.1.110 -> 67.9.155.15 ICMP 42 Echo (ping) request  id=0x65df, seq=0/0, ttl=58
3.731091 192.168.1.110 -> 67.9.155.16 ICMP 42 Echo (ping) request  id=0x6cf1, seq=0/0, ttl=43
3.731260 192.168.1.110 -> 67.9.155.17 ICMP 42 Echo (ping) request  id=0x2230, seq=0/0, ttl=52
3.731672 192.168.1.110 -> 67.9.155.18 ICMP 42 Echo (ping) request  id=0x5b79, seq=0/0, ttl=48
3.732010 192.168.1.110 -> 67.9.155.19 ICMP 42 Echo (ping) request  id=0xea59, seq=0/0, ttl=45
3.732192 192.168.1.110 -> 67.9.155.20 ICMP 42 Echo (ping) request  id=0xe1e6, seq=0/0, ttl=39
3.732437 192.168.1.110 -> 67.9.155.21 ICMP 42 Echo (ping) request  id=0xab55, seq=0/0, ttl=44
3.732800 192.168.1.110 -> 67.9.155.22 ICMP 42 Echo (ping) request  id=0xbd48, seq=0/0, ttl=47
```

Source: Wireshark

Figure 3-12 An automated ping sweep

Examining Abnormal Network Traffic Signatures, Part 3

- Port Scans
  - Attempt to connect to a computer's ports to see whether any are active and listening
    - An attacker who finds an open port can exploit any known vulnerabilities associated with any service that runs on that port
  - Signature of a port scan typically includes a SYN packet sent to each port on an IP address

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An Automated Port Scan

0.048185	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 1	[SYN] Seq=0 Win=3072 Len=0 MSS=1460
0.048323	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 2	[SYN] Seq=0 Win=4096 Len=0 MSS=1460
0.048405	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 3	[SYN] Seq=0 Win=3072 Len=0 MSS=1460
0.048484	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 4	[SYN] Seq=0 Win=3072 Len=0 MSS=1460
0.048563	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 5	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.048643	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 6	[SYN] Seq=0 Win=3072 Len=0 MSS=1460
0.048723	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 7	[SYN] Seq=0 Win=3072 Len=0 MSS=1460
0.048801	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 8	[SYN] Seq=0 Win=4096 Len=0 MSS=1460
0.048882	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 9	[SYN] Seq=0 Win=2048 Len=0 MSS=1460
0.049164	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 10	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.049593	192.168.1.110	->	192.168.1.132	TCP	54	45571 > 11	[SYN] Seq=1 Win=1024 Len=0 MSS=1460
0.054744	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 12	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.054841	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 13	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.054923	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 14	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.055003	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 15	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.056392	192.168.1.110	->	192.168.1.132	TCP	54	45571 > 16	[SYN] Seq=0 Win=2048 Len=0 MSS=1460
0.084972	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 17	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.085140	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 18	[SYN] Seq=0 Win=4096 Len=0 MSS=1460
0.085241	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 19	[SYN] Seq=0 Win=3072 Len=0 MSS=1460
0.085340	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 20	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.085437	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 21	[SYN] Seq=0 Win=2048 Len=0 MSS=1460
0.085539	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 22	[SYN] Seq=0 Win=2048 Len=0 MSS=1460
0.085636	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 23	[SYN] Seq=0 Win=3072 Len=0 MSS=1460
0.085734	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 24	[SYN] Seq=0 Win=3072 Len=0 MSS=1460
0.100816	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 25	[SYN] Seq=0 Win=2048 Len=0 MSS=1460
0.100947	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 26	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.101018	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 27	[SYN] Seq=0 Win=2048 Len=0 MSS=1460
0.101086	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 21	[SYN] Seq=0 Win=2048 Len=0 MSS=1460
0.101154	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 28	[SYN] Seq=0 Win=4096 Len=0 MSS=1460
0.101222	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 29	[SYN] Seq=0 Win=2048 Len=0 MSS=1460
0.101290	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 30	[SYN] Seq=0 Win=1024 Len=0 MSS=1460
0.101357	192.168.1.110	->	192.168.1.132	TCP	58	45571 > 31	[SYN] Seq=0 Win=1024 Len=0 MSS=1460

Source: Wireshark

Figure 3-13 An automated port scan

Examining Abnormal Network Traffic Signatures, Part 4

- Random Back Door Scans
  - Back door – an undocumented or unauthorized hidden opening (such as a port) through which an attacker can access a computer, program, or other resource
  - Probes a computer to see if any ports are open and listening that are used by well-known Trojan programs
  - Trojan programs
    - Applications that seem to be harmless but can cause harm to a computer or its files

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Examining Abnormal Network Traffic Signatures, Part 5

- Specific Trojan Scans
  - Vanilla scan – all ports from 0 to 65,535 are probed one after another
  - Strobe scan – scans only ports that are commonly used by specific programs
    - A common type of strobe scan searches IP addresses for the presence of a specific Trojan program
    - If a Trojan program has already operating, attackers save themselves the time of installing an new Trojan program

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Examples of Trojan Programs and Ports

Trojan	Protocol	Port
Trojan.Asprox	TCP	80, 82
W32.Spybot.pen	TCP	8076
W32.mytob.jw@mm	TCP	10027
Trojan.Mitglieder.h	TCP	14247
Sub-7 2.1	TCP	27573
Remote Windows Shutdown	TCP	53001
Back Office 2000	UDP	54321

Table 3-2 Examples of Trojan programs and ports

A Scan of a Single Host for Existing Trojans

	Scanner		Target				
20.929457	192.168.1.110	->	192.168.1.132	TCP	62	27634 > 4646	[SYN] Seq=0 Wi
20.929648	192.168.1.110	->	192.168.1.132	TCP	62	57801 > 5401	[SYN] Seq=0 Wi
20.929721	192.168.1.132	->	192.168.1.110	TCP	60	4646 > 27634	[RST, ACK] Seq=
20.929809	192.168.1.110	->	192.168.1.132	TCP	62	36559 > 4512	[SYN] Seq=0 Wi
20.929872	192.168.1.132	->	192.168.1.110	TCP	60	5401 > 57801	[RST, ACK] Seq=
20.929934	192.168.1.132	->	192.168.1.110	TCP	60	4512 > 36559	[RST, ACK] Seq=
20.929995	192.168.1.110	->	192.168.1.132	TCP	62	38739 > 3385	[SYN] Seq=0 Wi
20.930168	192.168.1.132	->	192.168.1.110	TCP	60	3385 > 38739	[RST, ACK] Seq=

Source: Wireshark

Figure 3-14 A scan of a single host for existing Trojans

Examining Abnormal Network Traffic Signatures, Part 6

- Nmap Scans
  - Network mapper (Nmap)
    - Popular software tool for scanning networks
  - Examples of Nmap scans
    - SYN scan – a progression of packets with only the SYN flag set
    - FIN scan –only packets with the FIN flag set
    - ACK scan –only packets with the ACK flag set
    - Null scan – sequence of packets that have no flags set
    - Xmas scan – sequence of packets that have the FIN PSH URG flags set

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Nmap SYN Scan

```
                Scanner            Target            Source  Destination
                                                      Port     Port
  3.805948 192.168.1.195 -> 192.168.1.194 TCP 58 47741 > 5631 [SYN] Seq=0 Win=4096 Len=0 MSS=1460

0000  00 25 9c 9e bc 0a 08 00 27 17 25 16 08 00 45 00   .%......'.%...E.
0010  00 2c 88 55 00 00 3b 06 72 a1 c0 a8 01 c3 c0 a8   .,.U..;.r.......
0020  01 c2 ba 7d 15 ff ec 8c a5 e7 00 00 00 00 60 02   ...}..........`.
0030  10 00 a0 5f 00 00 02 04 05 b4                     ..._......

  3.806419 192.168.1.195 -> 192.168.1.194 TCP 58 47741 > 2998 [SYN] Seq=0 Win=4096 Len=0 MSS=1460

0000  00 25 9c 9e bc 0a 08 00 27 17 25 16 08 00 45 00   .%......'.%...E.
0010  00 2c 57 63 00 00 33 06 ab 93 c0 a8 01 c3 c0 a8   .,Wc..3.........
0020  01 c2 ba 7d 0b b6 ec 8c a5 e7 00 00 00 00 60 02   ...}..........`.
0030  10 00 aa a8 00 00 02 04 05 b4                     ..........

  3.806908 192.168.1.195 -> 192.168.1.194 TCP 58 47741 > 9968 [SYN] Seq=0 Win=4096 Len=0 MSS=1460

0000  00 25 9c 9e bc 0a 08 00 27 17 25 16 08 00 45 00   .%......'.%...E.
0010  00 2c da 92 00 00 3b 06 20 64 c0 a8 01 c3 c0 a8   .,....;. d......
0020  01 c2 ba 7d 26 f0 ec 8c a5 e7 00 00 00 00 60 02   ...}&.........`.
0030  10 00 8f 6e 00 00 02 04 05 b4                     ...n......

  3.807382 192.168.1.195 -> 192.168.1.194 TCP 58 47741 > 5963 [SYN] Seq=0 Win=4096 Len=0 MSS=1460

0000  00 25 9c 9e bc 0a 08 00 27 17 25 16 08 00 45 00   .%......'.%...E.
0010  00 2c 56 0a 00 00 3b 06 a4 ec c0 a8 01 c3 c0 a8   .,V...;.........
0020  01 c2 ba 7d 17 4b ec 8c a5 e7 00 00 00 00 60 02   ...}.K........`.
0030  10 00 9f 13 00 00 02 04 05 b4                     ..........

  3.808002 192.168.1.195 -> 192.168.1.194 TCP 58 47741 > 1000 [SYN] Seq=0 Win=3072 Len=0 MSS=1460

0000  00 25 9c 9e bc 0a 08 00 27 17 25 16 08 00 45 00   .%......'.%...E.
0010  00 2c 9e 04 00 00 2a 06 6d f2 c0 a8 01 c3 c0 a8   .,....*.m.......
0020  01 c2 ba 7d 03 e8 ec 8c a5 e7 00 00 00 00 60 02   ...}..........`.
0030  0c 00 b6 76 00 00 02 04 05 b4                     ...v......
```

Source: Nmap

Figure 3-15 Nmap SYN scan

Nmap Xmas Scan

```
6.420376 192.168.1.195 -> 192.168.1.1   TCP 54 35048 > 6389 [FIN, PSH, URG] Seq=1 Win=1024 Urg=0 Len=0

0000  30 46 9a 15 33 bb 08 00 27 17 25 16 08 00 45 00   0F..3...'.%...E.
0010  00 28 f2 46 00 00 30 06 14 75 c0 a8 01 c3 c0 a8   .(.F..0..u......
0020  01 01 88 e8 18 f5 af 53 4d 2d 00 00 00 00 50 29   .......SM-....P)
0030  04 00 89 48 00 00                                 ...H..

6.420999 192.168.1.195 -> 192.168.1.194 TCP 54 35048 > 8651 [FIN, PSH, URG] Seq=1 Win=2048 Urg=0 Len=0

0000  00 25 9c 9e bc 0a 08 00 27 17 25 16 08 00 45 00   .%......'.%...E.
0010  00 28 44 01 00 00 2d 06 c4 f9 c0 a8 01 c3 c0 a8   .(D...-.........
0020  01 c2 88 e8 21 cb af 53 4d 2d 00 00 00 00 50 29   ....!..SM-....P)
0030  08 00 7b b1 00 00                                 ..{...

6.421224 192.168.1.195 -> 192.168.1.1   TCP 54 35048 > 2004 [FIN, PSH, URG] Seq=1 Win=4096 Urg=0 Len=0

0000  30 46 9a 15 33 bb 08 00 27 17 25 16 08 00 45 00   0F..3...'.%...E.
0010  00 28 90 98 00 00 37 06 6f 23 c0 a8 01 c3 c0 a8   .(....7.o#......
0020  01 01 88 e8 07 d4 af 53 4d 2d 00 00 00 00 50 29   .......SM-....P)
0030  10 00 8e 69 00 00                                 ...i..

6.421508 192.168.1.195 -> 192.168.1.194 TCP 54 35048 > 6389 [FIN, PSH, URG] Seq=1 Win=3072 Urg=0 Len=0

0000  00 25 9c 9e bc 0a 08 00 27 17 25 16 08 00 45 00   .%......'.%...E.
0010  00 28 34 b5 00 00 2a 06 d7 45 c0 a8 01 c3 c0 a8   .(4...*..E......
0020  01 c2 88 e8 18 f5 af 53 4d 2d 00 00 00 00 50 29   .......SM-....P)
0030  0c 00 80 87 00 00                                 ......

6.421803 192.168.1.195 -> 192.168.1.1   TCP 54 35048 > 1054 [FIN, PSH, URG] Seq=1 Win=3072 Urg=0 Len=0

0000  30 46 9a 15 33 bb 08 00 27 17 25 16 08 00 45 00   0F..3...'.%...E.
0010  00 28 5d 2e 00 00 3a 06 9f 8d c0 a8 01 c3 c0 a8   .(]...:.........
0020  01 01 88 e8 04 1e af 53 4d 2d 00 00 00 00 50 29   .......SM-....P)
0030  0c 00 96 1f 00 00                                 ......
```

Source: Nmap

Figure 3-16 Nmap Xmas scan

Identifying Suspicious Events

- Attackers often avoid launching well-known attacks
  - Use waiting intervals to fool detection systems
  - Scan throttling – often used by attackers to delay the progression of a scan over hours, days, or weeks
- Reviewing log files manually can be overwhelming
  - Must check them and identify potential attacks
- An IDPS can help you with this task
  - IDPSs depend on extensive databases of attack signatures

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Packet Header Discrepancies, Part 1

- Falsified IP address
  - Attacker can insert a false address into the IP header
    - Make the packet more difficult to trace back
  - Also known as IP spoofing
  - A land attack is an example
    - Occurs when a detected IP packet the same source and destination IP address
  - Localhost source spoof is another example
    - If source address of 127.0.0.1 occurs in a packet
- Falsified port number or protocol
  - Protocol numbers can also be altered
    - Port numbers should never be set to 0

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Packet Header Discrepancies, Part 2

- Illegal TCP flags
  - Look at the TCP flags for violations of normal usage
  - Examples of SYN and FIN flags misuse
    - SYN/FIN flags should not exist in normal traffic
    - SYN/FIN/PSH,SYN/FIN/RST,SYN/FIN/RST/PSH
      - Use is sometimes called an Xmas attack
    - Packets should never contain a FIN flag by itself
    - A SYN-only packet should not contain any data

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Packet Header Discrepancies, Part 3

- TCP or IP options
  - TCP options can alert you of an attack
    - Only one MSS or window option should appear in a packet
    - MSS, NOP, and SackOK should appear only in packets that have the SYN and/or ACK flag set
  - IP options
    - Originally intended as ways to insert special handling instructions into packets
    - Attackers mostly use IP options now for attack attempts
    - IPv6 removed options field and replaced it with extension headers

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Packet Header Discrepancies, Part 4

- Fragmentation abuses
  - Maximum transmit unit (MTU)
    - Maximum packet size that can be transmitted over a network
  - Packets larger than the MTU must be fragmented
    - Broken into multiple segments small enough for the network to handle
  - An IDPS should be configured to send an alarm if it encounters a large number of fragmented packets

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Packet Header Discrepancies, Part 5

- Fragmentation abuses (cont'd)
  - IPv4
    - Overlapping fragments – two fragments of the same packet have the same position within the packet
    - Fragments that are too large – IP packet can be no larger than 65,535 bytes
    - Fragments overwrite data – early fragments are transmitted along with random data and later fragments overwrite the random data
    - Fragments are too small – if any fragment (other than the final fragment) is less than 400 bytes, it has probably been crafted intentionally

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Packet Header Discrepancies, Part 6

- Fragmentation abuses (cont'd)
  - IPv6
    - Fragments with a destination address of a network device – if a router, firewall, or other device is the destination of fragmented IPv6 packets, a DoS attack might be intended
    - Fragments are too small - if any fragment (other than the final fragment) is less than 1280 bytes, it has probably been crafted intentionally
    - Fragments that arrive too slowly – fragments that take more than 60 seconds to deliver should be dropped

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Advanced Attacks, Part 1

- Advanced IDPS evasion techniques
  - Polymorphic buffer overflow attack
    - Uses a tool called ADMutate
    - Alters an attack's shell code to differ from the known signature many IDPSs use
    - Once packets reach the target, they reassemble into original form
  - Path obfuscation
    - Directory path in payload is obfuscated by using multiple forward slashes

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Advanced Attacks, Part 2

- Advanced IDPS evasion techniques (cont'd)
  - Common Gateway Interface (CGI) scripts
    - Scripts used to process data submitted over the Internet
    - Examples
      - Count.cgi
      - FormMail
      - AnyForm
      - Php.cgi
      - TextCounter
      - GuestBook

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Advanced Attacks, Part 3

- Advanced IDPS evasion techniques (cont'd)
  - Packet injection
    - Attackers can craft packets that comply with protocols that can be inserted into network traffic
    - Tools such as Nemesis are supposed to be useful for testing IDPSs and firewalls
      - Can be used to disrupt communications, spoof a variety of systems, and carry out a number of attacks

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Remote Procedure Calls, Part 1

- Remote Procedure Call (RPC)
  - Standard set of communication rules
  - Allows one computer to request a service from another computer on a network
- Portmapper
  - Maintains a record of each remotely accessible program and the port it uses
  - Converts RPC program numbers into TCP/IP port numbers

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Remote Procedure Calls, Part 2

- RPC-related events that should trigger IDPS alarms:
  - RPC dump
    - Targeted host receives an RPC dump request
  - RPC set spoof
    - Targeted host receives an RPC set request from a source IP address of 127.0.0.1
  - RPC NFS sweep
    - Targeted host receives series of requests for the Network File System (NFS) on different ports

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Summary, Part 1

- Common Vulnerabilities and Exposures (CVE)
  - Enables security devices to share attack signatures and information about network vulnerabilities
- Interpreting network traffic signatures can help prevent network intrusions
- Analysis of traffic signatures is an integral aspect of intrusion prevention
  - Possible intrusions are marked by invalid settings
- TCP flags are used in sequence to create a normal three-way handshake between two computers

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Summary, Part 2

- Learn what normal traffic signatures look like
  - Help identify signatures of suspicious connection attempts
- Suspicious network events
  - “Orphaned” packets
  - Land attacks
  - Localhost source spoof
  - Falsified protocol numbers
  - Illegal combinations of TCP flags

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Summary, Part 3

- Advanced attacks
  - Difficult to detect without a database of intrusion signatures or user behaviors
- Advanced attack methods include
  - Exploiting CGI vulnerabilities
  - Misusing Remote Procedure Calls

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